

Table 1: Representative list of mutators revealed more than three crashes from IssueMut. ★ =C23 standard

(At)=Attribute (BF)=Builtin function (Uo)=Unary Operator (FD)=Function Declaration (L)=Literal (I)=Initialization
 (P)=Parameter (E)=Expression (Sc)=Storage Class Specifier (S)=Statement (T)=Type (VD)=Variable Declaration
 (C)=Character

Id	Src.	Action	Pgm. Elements	Description
M1	G	Add	(BF)(E)	Adds built-in function <code>__builtin_assoc_barrier()</code> around the return expression.
M2	G	Swap	(E)	Swaps two arguments of function calls.
M3	G	Modify	(Sc)	Replaces <code>extern</code> storage class specifier with <code>static</code> .
M4	G	Remove	(Uo)	Removes a type cast operator.
M5	G	Modify	(P)(T)	Replaces const pointer parameters with non-const non-pointer parameters.
M6	G	Modify	(E)	Replaces a variable reference with a call expression.
M7	L	Remove	(VD)	Removes the size expression from an array variable declaration.
M8	L	Modify	(L)	★ Replaces integer literals with a large numeric value using a new integer literal suffix. (e.g., changing 123 to 66666...wb (The wb suffix is a bit-precise integer literal suffix introduced in the C23 standard)).
M9	L	Modify	(L)	★ Replaces integer zero literals into binary zero literals, which is introduced in the C23 standard (e.g., 0b0).
M10	L	Add	(At)(FD)	Adds <code>__attribute__((target_clones("default,avx")))</code> to function declarators.
M11	G	Modify	(L)	★ Replaces integer literals with the <code>nullptr</code> , a new null pointer constant introduced in C23.
M12	G	Modify	(S)	Replaces standard goto statement with <code>asm goto</code> equivalents.
M13	G	Modify	(S)(VD)	Replaces inline assembly register constraints of <code>asm stmt</code> with floating-point constraints (e.g., changing +r to +d). Replaces <code>'__asm__'</code> with <code>'__asm'</code> in variable declarations.
M14	G	Add	(S)	Adds a conditional assignment statement to an array element.
M15	G	Modify	(S)	Replaces <code>for</code> loop constructs with equivalent <code>while</code> loop implementations.
M16	G	Modify	(FD)	Replaces function identifiers with generic identifier.
M17	G	Add	(FD)	Adds explicit <code>extern</code> declarations for implicitly declared functions in call expressions.
M18	G	Modify	(VD)	Removes temporary variables and replaces references with their initializers.
M19	G	Add	(At)(FD)	Adds <code>__attribute__((noreturn))</code> to function declarations.
M20	G	Remove	(At)	Removes the attribute specifier from a function declarator or a variable declaration.
M21	G	Modify	(E)	Replaces type cast expressions in return statements with equivalent variable declarations and assignments.
M22	G	Modify	(P)(T)	Replaces function parameter types with <code>char</code> .
M23	G	Add	(T)	Modifies a struct by introducing a typedef alias.
M24	G	Add	(P)	Adds two <code>long long</code> type parameters to a function declarator.
M25	G	Add	(BF)(E)	Adds <code>__builtin_assume_aligned</code> to destination pointers in calls with <code>__builtin_memcpy</code> .
M26	G	Remove	(I)	Removes the zero initialization expression from a pointer variable declaration.
M27	G	Remove	(I)	Removes the initialization expression from a variable declaration.
M28	G	Add	(T)	Adds a field of type <code>long double</code> to a union.
M29	L	Add	(At)	Adds the <code>no-</code> prefix to existing target attribute specifier.
M30	L	Modify	(T)	Replaces platform-specific types declarations with explicit-size alternatives. (e.g., <code>size_t</code> with <code>long long</code>).
M31	L	Remove	(Uo)	★ Removes <code>typeof()</code> operators, introduced in the C23 standard.
M32	L	Add	(C)	Adds escape characters (e.g., backslashes) after line break characters.
M33	L	Remove	(P)	Removes void parameter from function declarator.